



TRUSTLINE

A Self-Hosted End-to-End Encrypted Real-Time Collaboration Platform
for Confidential Enterprise Workflows

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Abstract

Addressing vulnerabilities in modern enterprise collaboration.

- The problem of centralized trust risks and cloud service provider vulnerabilities.
- Elimination of implicit trust models to protect proprietary data.
- Implementation of strict client-side Zero-Knowledge Architecture.
- Real-time encrypted synchronization using Conflict-Free Replicated Data Types (CRDTs).
- Self-hosted deployment ensuring absolute data sovereignty for enterprises.
- Objective: Highly secure, high-performance, and seamless confidential workflows.



Abstract (Continued)

Core technologies driving Trustline's secure architecture.

- X3DH & Double Ratchet: Advanced End-to-End Encryption ensuring forward and post-compromise secrecy.
- Device Identity Signature: Cryptographic passwordless authentication eliminating credential theft.
- SQLCipher & Argon2id: Memory-hard local vault encryption providing extreme brute-force resistance.
- Blind Router & PostgreSQL RLS: Stateless message relaying with strict tenant-level isolation.

Synchronization Latency: ~110 ms

Authentication Latency: ~359 ms



Literature Survey

Author (Year)	Title	Technique used	Merits	Demerits
Signal Protocol	X3DH & Double Ratchet	Strong E2EE, Forward Secrecy	Poor group scaling, tied to phone numbers	
Matrix Protocol	Decentralized Federated Sync	Open standard, self-hostable	Metadata leakage, complex key management	
Yjs (CRDTs)	Distributed mathematical convergence	No central ordering server needed	State bloat over time	
Zero Trust Arch.	Continuous verification	Mitigates lateral movement	Often lacks strong data encryption at rest	
Device Signatures	Cryptographic challenge-response	Phishing resistant	Device recovery complexity	



Identified Research Gaps

Author (Year)	Title	Technique used	Merits	Demerits



Problem Statement

The inherent insecurity of centralized enterprise tools.

- Implicit Trust & Lateral Movement: Traditional firewalls fail when the network is breached.
- Cloud Privacy Risks: Providers holding decryption keys become high-value targets.
- Insider Threats: Administrators have unfettered access to plain-text databases.
- Metadata Exposure: Traffic analysis profiles users even when payload data is encrypted.
- Authentication Weaknesses: Passwords and legacy 2FA are vulnerable to phishing and GPU cracking.
- Enterprise Sovereignty: Organizations lose absolute control over their sensitive IP.



Objectives of the Study

The foundational goals of the Trustline platform.

- Implement true Zero-Knowledge communication where servers cannot read data.
- Develop a "Blind Routing" backend that relays encrypted packets without inspecting content.
- Integrate highly secure, cryptographic challenge-response authentication avoiding passwords.
- Enable encrypted real-time synchronization without relying on a central authority.
- Guarantee forward secrecy and post-compromise security for all communication.
- Provide a fully self-hosted infrastructure for complete enterprise control.



Proposed Overall Architecture



Module 1 - Cryptographic Key Exchange



Module 1 - Double Ratchet Protocol

Continuous security for ongoing communication sessions.

- Root Chain Synchronization: Orchestrating the derivation of all subsequent key chains.
- Sending/Receiving Chains: Independent key rotation for inbound and outbound message flows.
- Message Key Derivation: Every single message is encrypted with a unique, one-time key.
- Forward Secrecy: Compromised current keys cannot decrypt previously sent messages.
- Post-Compromise Security: A compromised session heals itself upon the next asynchronous key exchange.



Module 2 - Blind Router Backend



Module 2 - Enterprise Data Segregation

Enforcing strict tenant boundaries using PostgreSQL.

- Tenant Isolation: Complete logical separation of data between different enterprise organizations.
- Org-Level Access Control: Restricting data access rigorously based on organization IDs.
- Secure Row-Level Security (RLS): Database-level policies enforcing isolation independently of application logic.
- Attack Mitigation: Prevents cross-tenant data leakage even in the event of an application-layer vulnerability.



Module 3 - Cryptographic Authentication



Module 3 - Secure Local Persistence

Protecting data at rest on client devices.

- Secure Key Derivation: Utilizing Argon2id for deriving robust encryption keys from user credentials.
- Memory-Hard Function: Defending against highly parallelized GPU/ASIC cracking arrays.
- Vault Encryption: Encrypting the entire local SQLite database using SQLCipher.
- Rapid Indexed Search: Implementing FTS5 over decrypted data in memory for instant query responses.



Module 4 - Distributed Encrypted Synchronization



Module 4 - Device & Vault Management

Enterprise control over edge devices.

- Secure Device Onboarding: Cryptographic verification required when adding new devices to an account.
- Encrypted Storage Management: Local handling of cryptographic materials and encrypted caches.
- Remote Wipe Capabilities: Administrators can securely invalidate device sessions and trigger local vault destruction.
- Shamir's Secret Sharing: Securely partitioning recovery keys to prevent single-point compromise.



Experimental Results & Benchmarks



Hardware Requirements

System specifications for deploying and running Trustline.



Software Requirements

The technological stack powering the platform.



Infrastructure Stack Summary



Data Collection & Testing Simulation



Data Characteristics



Module 1 Results - X3DH



Module 1 Results - Double Ratchet



Module 2 Results - Blind Routing



Module 2 Results - Tenant Isolation



Module 3 Results - Authentication



Module 3 Results - Cryptographic Vault

Validation of local storage security.

- Argon2id Performance: Key derivation tuned for client hardware, balancing security against login delays.
- Secure Vault Encryption: SQLCipher maintained strict integrity over the local SQLite database.
- Brute-Force Resistance: Validated against massive dictionary attacks, projecting over 100,000 years to compromise a secure vault.



Module 4 Results - CRDT Sync

Validation of real-time distributed consistency.

- Synchronization Latency: Maintained ultra-low latency (~110ms) across geographically distributed clients.
- Encrypted Collaboration Results: Simultaneous editing functioned perfectly over the encrypted WebSocket tunnel.
- Distributed Convergence Validation: Clients recovering from offline states successfully mathematically merged all pending updates flawlessly.



Module 4 Results - Device Management

Validation of administrative edge control.

- Secure Multi-Device Sync: Flawless propagation of application state across desktop and mobile clients.
- Secure Device Wipe Results: Remote revocation successfully destroyed local cryptographic materials instantaneously.
- Vault Recovery Validation: Shamir's Secret Sharing successfully reconstructed vault access during simulated master device loss.



Conclusion

Summary of project achievements.

- Validated the feasibility of high-performance, strictly Zero-Knowledge enterprise collaboration.
- Proved that advanced cryptography (AES-GCM, Argon2id) does not hinder sub-250ms real-time synchronization constraints.
- Successfully replaced weak password paradigms with highly secure, cryptographic challenge-response authentication.
- Delivered a fully functional, self-hosted architecture guaranteeing digital sovereignty for enterprises.
- Effectively protected extremely confidential workflows from lateral movement, cloud breaches, and insider threats.



Future Scope



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